



FROM KHAVDA TO KAMUTHI: ADANI GREEN ENERGY SCALES NEW HEIGHTS IN CLEAN POWER

From pioneering wind-solar hybrid energy projects to building the world's largest renewable plant, the Adani Group's renewables capacity push is rewriting the rules of sustainable infrastructure at scale

Adani Green Energy Limited (AGEL) has emerged as a towering force in the global renewable energy sector. As India's largest renewable energy company and one of the world's leading green power producers, AGEL's rise reflects a bold vision brought to life with extraordinary speed and precision by the diversified Adani Group, an integrated infrastructure major spanning ports, airports, logistics and energy. With an operational portfolio now exceeding 15,850 MW (megawatt), AGEL's footprint spans 12 Indian states, encompassing solar, wind and hybrid energy systems. In financial year (FY) 2025 alone, the company added 3,309 MW, setting a national record for the largest renewable capacity addition by a single entity in one year.

This impressive portfolio comprises over 11,000 MW of solar, 2,000 MW of wind and 2,560 MW of wind-solar hybrid capacity. The power generated is capable of lighting up approximately 8 million homes and can supply electricity to 13 Indian states individually. It can illuminate the entire Northeast region with renewable energy, which has an estimated cumulative population of around 52.1 million—up from approximately 46.5 million, as per the 2011 Census.

The sheer scale of AGEL's operations mirrors its commitment: to energise India's future without compromising its environment.

Innovation, Technology, and ESG Leadership

Beyond numbers, AGEL is reshaping the renewable energy narrative through relentless innovation and sustainability. In 2016, AGEL had commissioned India's largest single-site solar power plant in Kamuthi, Tamil Nadu, which set a benchmark in solar energy generation and operational efficiency. It didn't stop there. AGEL pioneered solar-wind hybrid energy projects to enable round-the-clock renewable power, addressing a major challenge in clean energy reliability. This capability is critical in ensuring that renewable energy can serve as a consistent and reliable source.

Today, the title of the world's largest single-site solar PV (photovoltaic) plant is held by Noor Abu Dhabi, but AGEL has already turned its focus to something even more ambitious: Khavda.

Khavda: The World's Largest Renewable Energy Project

Located in the sun-and-wind-rich expanse of Kutch, Gujarat,

AGEL's Khavda project is nothing short of revolutionary. Spread over 538 square kilometres—nearly five times the size of Paris—Khavda will generate 30 GW of clean power by 2029, making it the largest renewable energy installation in the world. As of mid-2025, AGEL has operationalised over 5.5 GW at Khavda. Once fully built, it will produce 87.41 billion units of clean electricity annually—enough to power entire nations such as Belgium, Chile or Switzerland. The energy generated will electrify 17.4 million homes, a number comparable to households in Poland, Canada or Spain.

Khavda is not just a technological feat; it is a green employment engine, set to create over 15,000 direct and indirect jobs. It reflects AGEL's deep integration of social impact within its growth model.

The plant features bifacial solar modules, which absorb sunlight from both sides for higher efficiency, and waterless robotic cleaning systems, critical in the arid region where water conservation is paramount. It also deploys 5.2 MW onshore wind turbines, one of the largest globally, optimised for the region's wind patterns. In Khavda, scale meets sustainability.

The company's strategic decisions—from site selection to supply chain design—reflect a commitment to balancing economic ambition with ecological responsibility. AGEL's growth is designed to last not just for the next quarter, but for the next generation.

AGEL and India's Clean Energy Future

India's clean energy transformation is not simply about meeting targets—it is about reimagining infrastructure, livelihoods, and national priorities. In this complex equation, AGEL is not just a contributor but a catalyst.

Unlike many players that retrofit sustainability into operations, AGEL embeds it into its DNA (Deoxyribonucleic Acid). The company's strategic decisions—from site selection to supply chain design—reflect a commitment to balancing economic ambition with ecological responsibility. AGEL's



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